Img16DatSer.exe
16-bit Headerless Images DataSeries viewer (for Windows, Linux+WinE)

(c) MIT License Sep-2026 by S.Bochkarev

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1. The introduction

1.1 The scope & purpose

This utility is 16-bit <little-endian> header-less image viewer for Windows and (Ubuntu24,26) WinE.
Typical files' extensions are:
- *.bin - some cameras "o naturel";
- *.raw - ImageJ, AdobePhotoshop, Aldus Photostyler (note that these programs vary big|little-endianess); some cameras
- *.img, *.dat - some programs and cameras

= The main functionality of this utility:
- read 16bit little-endian pixels from image_file,
- hold the pixels in memory as 32bit signed ints,
- displays those as 8bit grayscale.

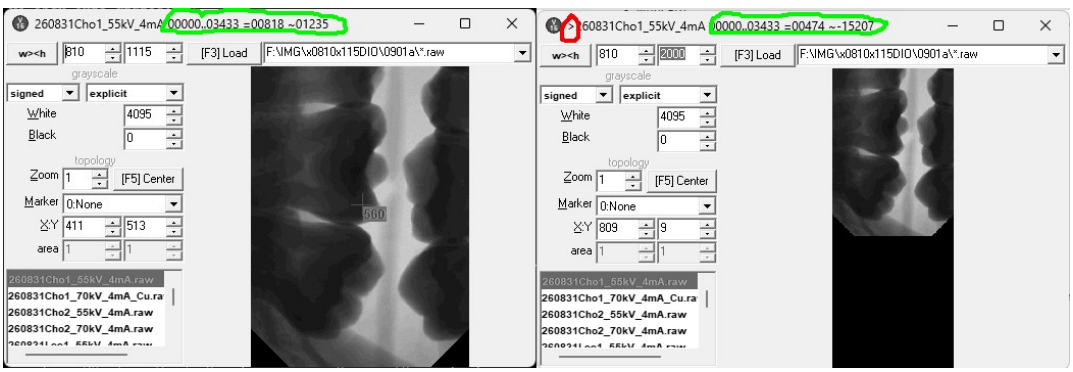
= Specific builds:
- "WindowsGDI" is default,
- "DirectX 2D" supports only "halftone bitblt-stretch"; this could become rational in ~ old Linux-PCs (when versatile WinE-GDI is too slow).

= Binary's name defines initial state of pixels' interpretation:
- default : - [signed],
- *U.exe, *u.exe: - [unsigned].

the input:
= section [View] in ini-profile
- this_exe_directory/%IniProfileOfSpecificSDK% (if one exists),
- this_exe_directory/this_exe_name.ini (otherwise);

= optional cmdline
- "w h wildcard",
- "wildcard",
where wildcard composes as "directory/fname_wildcard" (e.g. ~"256 256 *.dat", ~"640 480 z:/img/1????.bin", ~"/home/user/img/*.raw").

window-title:
- <prefix> file_name min..max =median ~average
- '?'="no image", >="[w]*[h]>file_size",...



1.2 The development tools

= The projects are :
- EProg2008.vcproj : VC- build-toolchain (MS VS 2008)
- EProg.vcxproj : MS- build-toolchains (tested MS VS 2017,2002,2026)

= The character sets :
Assuming the WinE, build this project ~always as UNICODE. MBCS is possible for Windows (but ~has no purpose).

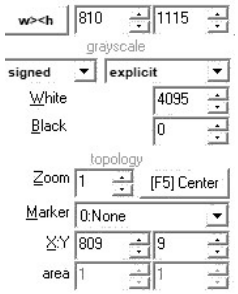
1.3 Usage of VSCode

= File .vscode/c_cpp_properties.json

```
// this is rather a stub to have minimalistic code-highlights in VSCode
// and one-click build (when started under VS-developer-comman-prompt)
{  "version": 4,
   "configurations": [
      { "name": "Win64",
        "includePath": ["${workspaceFolder}/."],
        "defines": ["_WIN32"],
        "compilerPath": "cl.exe",
        "cStandard": "c17",
        "cppStandard": "c++17",
        "intelliSenseMode": "windows-msvc-x64",
        "configurationProvider": "ms-vscode.cmake-tools",
        "browse": {
           "path": ["${workspaceFolder}/**"],
           "limitSymbolsToIncludedHeaders": true
        }
      }
   ]
}
```

= There is no known tactic to make a robust CMakeLists.txt for a typical MFC-project.

2. The main View



2.1 View-controls

```
= [w><h]         :   - swaps the width,height

= [Load]         :   - browse & load image_files

= [combo box] (interprets pixels in image_files)
  - signed       :   - default -32768..32767
  - unsigned     :   - 0..65535 (when named as "View16U.exe")

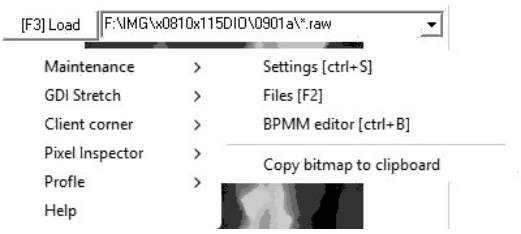
= [combo box] (gray_gamut)
  - explicit     :   - black=>RGB(0,0,0), white=>RGB(255,255,255)
  - THist        :   - cut 0.1% of pixels
  - inv THist    :   - cut 0.1% of pixels and invert

= [Zoom]         :   - magnifies frame's rectangle versus displayed one

= [combo Marker]:    - draws an accessory alongside image

= [edits X,Y]    :   - Marker's location

= [area]         :   - Marker's size
```



2.2 Pop-up menu

Appears by [F9], right mouse btn

```
      Maintenance
= the dialogs (show particular help by [F1])
  - Settings   :    - UI-settings,
  - Files      :    - operate with items of files' list,
  - BPMM editor:    - operate with BadPixels'Mask-overlays.

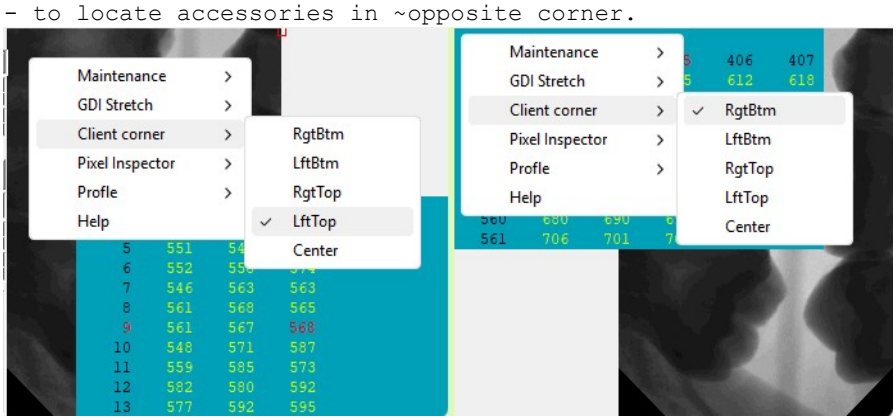
= the commands
  - Copy bitmap to clipboard,
  - Copy ROI text to clipboard,
  - Copy recent stats to clipboard.
  - [Execute]; the title might be 'Windows Explorer' or 'WinE Browser'.


      GDI-stretch selection <when allowed>
```

- Fast : - StretchBlt(...COLORONCOLOR,)
- Halftone : - ...HALFTONE (the only option in DirectX-build),
- White|Black details: - ...WHITEONBLACK|BLACKONWHITE.

Client corner's selection

- to locate the image (when aspect_ratio reduces client_area),
- to locate accessories in ~opposite corner.

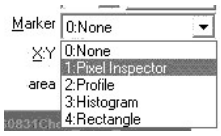


Pixel Inspector's selection

- 5x5 | 7x7 | etc.

Profile's selection

- Row | Column | Both.



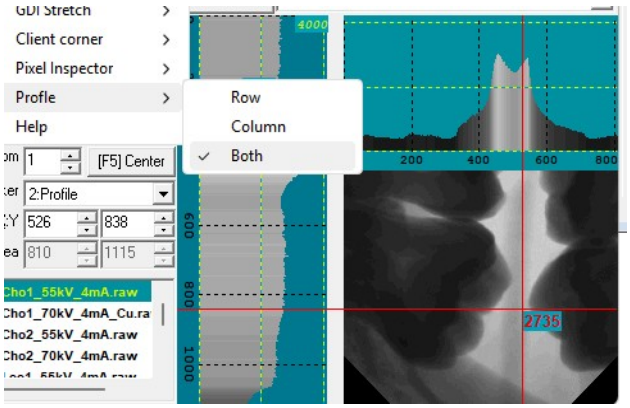
2.3 The markers

1:Pixel inspector

- ROI-center is view's spin-edits [X,Y]
- ROI-area is selection in pop-up menu

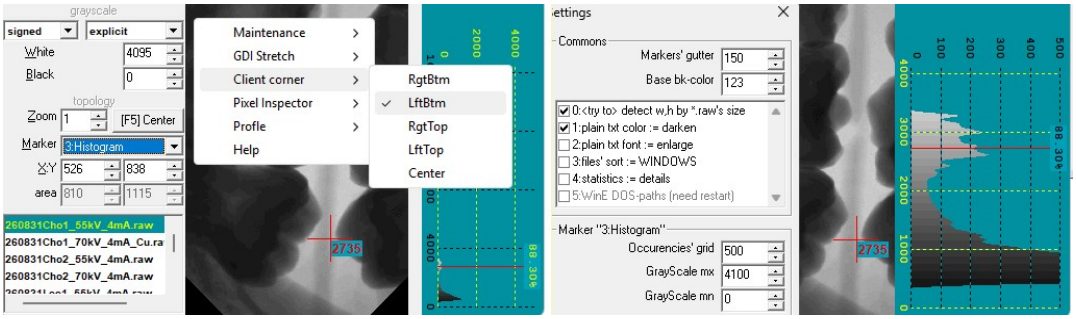
2:Profile

- the lines Row|Column|Both follow selection in the pop-up menu
- the grids follow the view



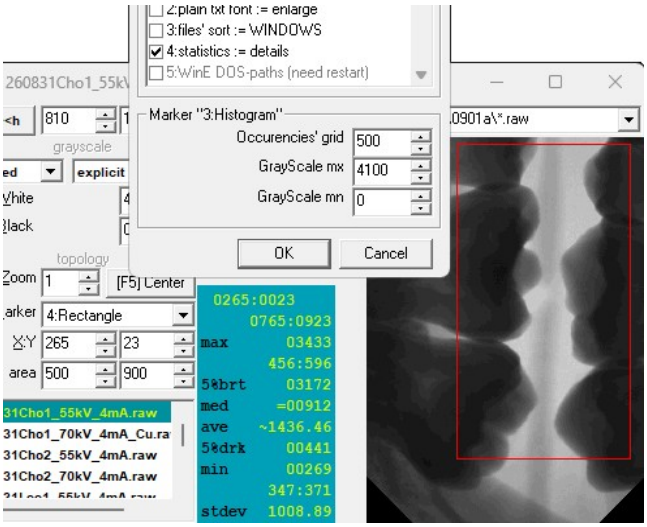
3:Histogram

- the grids follow [Settings]

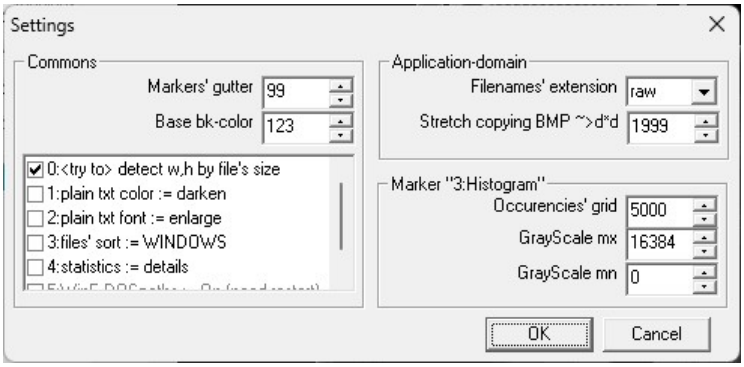


4:Rectangle

- ROI-lft_top is view's [XY]
- ROI-area is view's [area
- presenting statistics follow [Settings]



3. Modal dialog [Settings]



3.1 settings "Commons"

```
= Markers' gutter   ~50..200
  <in pixels> width of markers 2:Profile, 3:Histogram.

= Base bk-color     ~50..200
  <in LSB> brightness of drawing accessories.

= Bit_flags
  - 0:detect w,h by file's size" - affects [Load];
the order of checks:
 a) the same FileSize in stored wildcards,
 b) if FS mod sqrt(FS)=0 then w:=h:=sqrt(FS),
 c) if FS mod w=0 then h:=FS div w,
 d) if FS mod h=0 then w:=FS div h.
  -  :"plain txt ... := ..."      - affect drawing accessories,
  - 3:"files' sort := WINDOWS"    - default is alphabet_order,
  - 4:"statiscics := details"     - affects markers 3:Histogram, 4:Rectangle,
  - 5:"WinE DOS-paths := On"      - show DOS-paths, force WinExplorer's stub (affects
under WinE).
```

3.2 settings "Application-domain"

```
= Filenames' extension:
  - *.raw ~ ImageJ, AdobePhotoShop, medical camers
  - *.bin ~ industrial cameras
  - etc

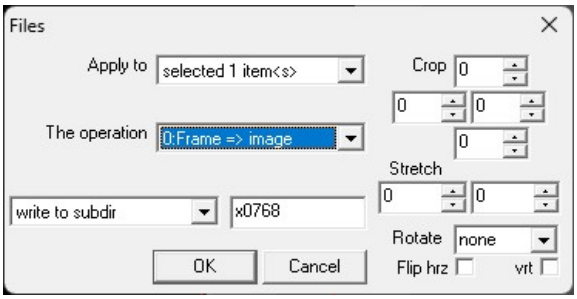
= Stretch copying BMP ~>d*d
  stretch CF_DIB for clipboard if exeeds this area (this mainly for WinE's
limitation). WinE might don't copy CF_DIB ~>4MB (i.e. set "d~<1999" for grayscale 8-
bit BMP in this application).
In Windows:
  - this adhoc limitation might be spractical;
  - one might assign large value (d~>9999) and copy large BMPs via clipboard exactly.
```

3.3 settings "marker 3:Histogram"

```
= Visible axes:
  - abscissa  - by "GrayScale_mn..mx",
  - ordinate  - by "0..Occurencies_grid".
```

4. Modal dialog [Files]

```
Terminology:
  - 'Frame' - inp image 'as is'
  - 'Image' - transformed 'Frame'.
```




```
      Apply to:
= current selection in files'list,
= whole files'list.

      The operations:
= 0:Frame => Image (in the following order):
  - Crop margins
  - Stretch to specified WxH
  - Rotate
  - Flip
  - Write ./SubDir/image_file or overwrite existing image_file

= 1:Write simple average frame '_ave'.

= 2:Frame => BMP (cut grayscale image_file<s> to 8-bit-gray *.BMP).

= 3:Delete file<s>.

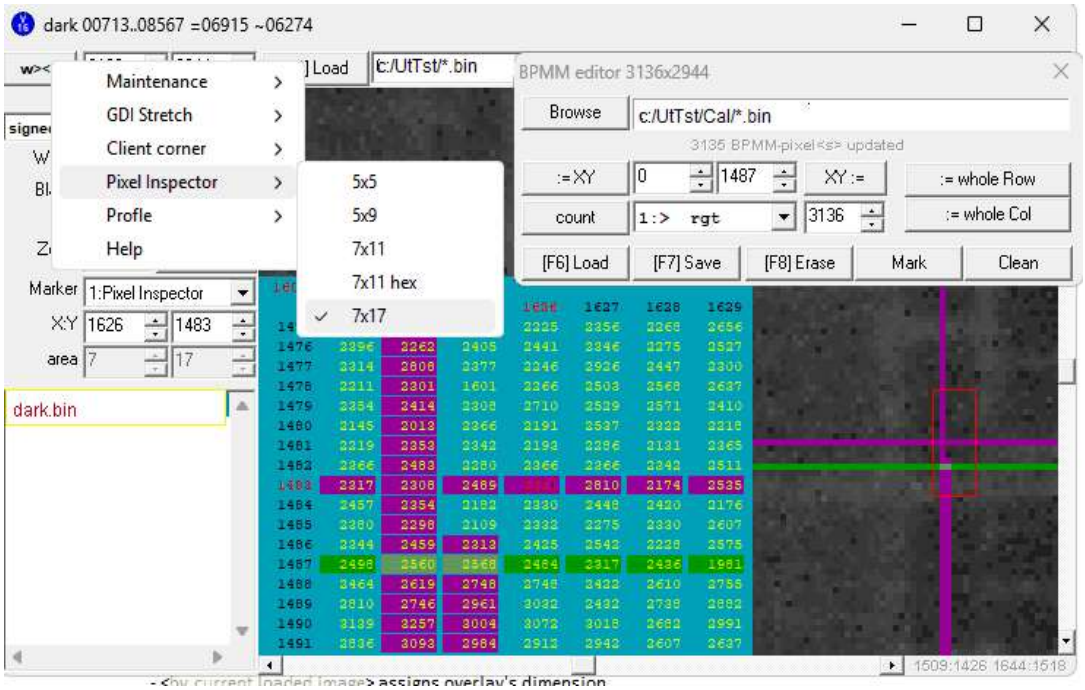
= 4:OS-specific 'open'
  - Invokes ~ShellExecute('open',image_file).

= 5,6:Hide <un>selected file_name<s> in the files'list
  This is to compare histograms, profiles, etc of two image_files, when those aren't
neighbors in files'list initially.
  - a) <by this operation> make them neighbors in the list,
  - b) <by arrow_keys up|down> move files'list_caret and watch quantitative changes
between two charts or between two image_files.") ); }
```

5. Dialog [BPMM-editor]

Infrared cameras, flat-panel X-ray -detectors, industrial optical cameras shall equalize individual $P_{ixelhoton}T_{ransfer}C_{urve}s$.
The PTC of aliased pixel can't be properly equalized (by it's currrent value and pre-captured calibration-data alone).
Then display of aliased pixel need a 2D-filtration (involving good neighbors). Pixel's alias could be:
= transient, where the fact of alias is detected (and then fixed) run-time;
= permanent, where the fact of alias is:
- detected automatically (from calibration data-series) in advance;
- marked manually in advance.

Respectivelly, calibration-data of those cameras <might> incapsulate two overlay-files:
- automatic BadPixelsMap,
- manual BadPixelsMap.



This dialog shows overlay **BadPixels' map** <automatic>. Shows & edits overlay **BadPixels' map manual**.

At open this dialog:
= <by current loaded image> assigns overlay's dimension,
= <tries to> loads the overlays from SDK-specific location.
= displays status line :
~ e.g. "BMP NNNNN pixels, BPMM doesn't exist",
~ e.g. "BMP doesn't exist, BPMM MMMMN pixels".

```
Editing area (marked by green color in UI-topology):
- BPMM_org    two spin_edits (x & y) specify 2D-point;
- BPMM_cnt    combo_box & spin_edit specify direction & count.

The actions:
[Browse]            selects location of loading overlays;
[:= XY]             BPMM_org := Marker_XY;
[XY :=]             Marker_XY := BPMM_org;
[count :=]          assigns BPMM_cnt by Marker_XY;
```

```
[:= whole Row]      assigns BPMM_org, BPMM_cnt by Marker_XY;
[:= whole Col]      -"-;
[Load]              loads the overlays (abandons changes);
[Save]              saves BPMM in file;
[Erase]             fills BPMM by 0 ([Erase]+[Save] delete BPMM.raw);
[Mark]              fills editing area by 1;
[Clean]             fills editing area by 0.
```

Pixels' colors and text-background:

```
- dark magenta  color -    <>0 in BPM,
- light magenta color -    <>0 in BPMM,
- green         color -    within "editing area".
```

To enhance visibility of ~pixel-size-details on specific monitor:

- utilize GDI-stretch selection ~= "*White|Black details*" in the pop-up menu (perhaps prefer GDI-build against GPU-build);
- adjust "base bk-color" in [Settings].